

Ê PHASOR TEMPLATES (Air: ε_r=μ_r=1)

Wavenumber per medium <i>nonmag.</i>			
$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c}$ (rad/m) $\Rightarrow k_2 = k_1\sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$			
Direction table $u \in \{x, y, z\}$ = axis perp. to boundary			
Inc dir	Inc	Refl	Trans
+û	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
-û	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. ê, amplitude a ₀):		
$\tilde{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u}$ (V/m)	(incident wave, Med 1)	
$\tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u}$ (V/m)	(reflected wave, Med 1)	
$\tilde{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u}$ (V/m)	(transmitted wave, Med 2)	
(lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$)		
Total in medium 1: $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$.		
ê by polarization plug into templates above		
Pol.	ê	Note
Lin. x̂	ŷ	E _y = 0
Lin. ŷ	ŷ	E _x = 0
Lin. at 45°	(x̂ + ŷ)/√2	in phase
RHC	x̂ - jŷ	δ = -π/2
LHC	x̂ + jŷ	δ = +π/2
General	a _x x̂ + a _y e ^{jδ} ŷ	elliptical

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

<i>Med. 1 (z < 0) incident side; Med. 2 (z ≥ 0) transmitted.</i>			
	⊥ pol (θ _i = 0)	pol	
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ rel.	τ = 1 + Γ	τ _⊥ = 1 + Γ _⊥	τ = (1 + Γ) $\frac{\cos \theta_i}{\cos \theta_t}$
R refl. pwr	Γ ²	Γ _⊥ ²	Γ ²
T trans. pwr	τ ² $\frac{\eta_1}{\eta_2}$	τ _⊥ ² $\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	τ ² $\frac{2 \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
R + T total	T = 1 - R	T _⊥ = 1 - R _⊥	T = 1 - R
Notes: sin θ _t = √μ ₁ ε ₁ /μ ₂ ε ₂ sin θ _i ; nonmag. ⇒ η ₂ /η ₁ = n ₁ /n ₂ .			
Intrinsic impedance η_i plug into table above			
$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)} \quad \eta_0 = 120\pi \approx 377 \Omega$			
Default μ _r = 1 (air, dielectric). Use μ _r ≠ 1 only if problem says "ferromagnetic"/iron/ferrite/etc.			
Low-loss correction $\eta_c \propto \sigma/(\omega\epsilon) \ll 1$			
$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$			
For Γ/τ just use η ₀ /√ε _r . The j term is for η _c , θ _η . See LOSSY MEDIA — 5 CASES.			

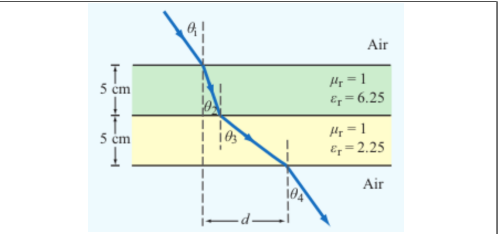
POWER REFLECTION & TRANSMISSION

% reflected <i>always</i> %P _r = 100 Γ ²
% transmitted <i>η-ratio matters!</i> %P _t = 100 τ ² η ₁ /η ₂
Check: %P _r + %P _t = 100.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary <i>angles from normal</i>	$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$
Multilayer chain <i>stack of slabs</i> 1 → 2 → 3 → 4	$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$
Air-slabs-air shortcut <i>outer media identical</i>	θ _{exit} = θ _{incident}
<i>Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).</i>	

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t_i, refracted angle inside slab i Slab-indexed θ_i = angle inside slab i; t_i = slab thickness (m)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

LOSSY MEDIA — 5 CASES

Loss tangent $\epsilon' = \epsilon_r \epsilon_0$; $\epsilon_0 = 8.854 \times 10^{-12}$ F/m		
$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$		
Type	σ/ωε	η _c , α, β
Lossless (σ = 0)	0	η = η ₀ /√ε _r exact α = 0, β = ω√ε _r /c
Low-loss	< 10 ⁻²	η _c ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma \epsilon_r}{2\omega \epsilon'}\right)$ α ≈ $\frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}$, β ≈ $\frac{\omega \sqrt{\epsilon_r}}{c}$
Quasi-cond.	10 ⁻² –10 ²	exact: (η/√X)e ^{jθ_η} see below
Good cond.	> 10 ²	η _c ≈ (1+j)√πfμ/σ α ≈ β ≈ √πfμσ
Magnetic	any	keep full √μ _r /ε _r η ₀

For Γ/τ: drop j correction; use η₀/√ε_r. Magnetic: keep μ_r.

Low-loss quick params σ/(ωε) ≪ 1, μ_r = 1
α ≈ $\frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}$ (Np/m), β ≈ $\frac{\omega \sqrt{\epsilon_r}}{c}$ (rad/m), η_c ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}}$ (Ω)
Exact (quasi-cond) X = √[1 + (ε''/ε')²]
α = ω√ $\frac{\mu \epsilon'}{2}$ √X-1, β = ω√ $\frac{\mu \epsilon'}{2}$ √X+1
θ_η = $\frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'}$

RECTANGULAR WAVEGUIDE — MODES

TE mode E_z = 0, H_z ≠ 0 TM mode H_z = 0, E_z ≠ 0
Dimensions a × b with a > b along x̂, ŷ; wave in ẑ.
E_x form (TE) read m, n from arguments
 $E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$
coef. on x = mπ/a; coef. on y = nπ/b.

Identify TE_{mn} vs TM_{mn} cos/sin pattern of E_x is identical for both!

1. Read the problem statement (e.g. “a TE wave” → TE).
2. If m = 0 or n = 0 in the mode label ⇒ must be TE (TM forbids 0).
3. Source field H_z(x, y) given ⇒ TE. Source field E_z(x, y) given ⇒ TM.

TE allows at least one of m, n ≥ 1 (other can be 0). TM needs both m, n ≥ 1. So TE₁₀ exists but TM₁₀ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

e^{-jβz} factor omitted. k_c² = (mπ/a)² + (nπ/b)². Ê in V/m, Ĥ in A/m, Z in Ω.

TE mode <i>generator: Ê_z</i>	
$\tilde{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{E}_z = 0, \tilde{H}_x = -\tilde{E}_y/Z_{TE}, \tilde{H}_y = \tilde{E}_x/Z_{TE}$	
$Z_{TE} = \eta / \sqrt{1 - (f_c/f)^2}$	
TM mode <i>generator: Ê_z</i>	
$\tilde{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{H}_z = 0, \tilde{H}_x = -\tilde{E}_y/Z_{TM}, \tilde{H}_y = \tilde{E}_x/Z_{TM}$	
$Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$	
TEM plane wave <i>for reference</i>	
$\tilde{E}_x = E_0 e^{-j\beta z}, \tilde{H}_y = \tilde{E}_x/\eta$	
$\eta = \sqrt{\mu/\epsilon} \text{ (}\Omega\text{)}, f_c \text{ N/A, } k = \omega\sqrt{\mu\epsilon}$	
Properties common to TE/TM	
$f_c = \frac{u_{p0}}{2} \sqrt{(m/a)^2 + (n/b)^2} \text{ (Hz)}$	
$\beta = k\sqrt{1 - (f_c/f)^2} \text{ (rad/m)}$	
$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_c/f)^2}} \text{ (m/s)}$	

CUTOFF FREQUENCY

Common cutoffs (a > b) f _c formula in Table 8-3	
Mode	f _{mn}
u _{p0}	c/√ε _r (hollow: u _{p0} = c)
TE ₁₀	f ₁₀ = u _{p0} /(2a) (dominant)
TE ₀₁	f ₀₁ = u _{p0} /(2b)
TE ₂₀	f ₂₀ = u _{p0} /a = 2f ₁₀
TE ₁₁ , TM ₁₁	f ₁₁ = $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$
Propagation requires f > f _{mn} . In Z _{TE} /Z _{TM} and u _g , f _c = f _{mn} of the excited mode.	

ε_r FROM DISPERSION

Given ω, β, mode (m, n) result is unitless	
$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$	(unitless)
Inputs: ω (rad/s), β (rad/m), a, b (m), c = 3 × 10 ⁸ (m/s), m, n (unitless).	

GROUP VELOCITY & TRAVEL TIME

$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2}$ (m/s), $u_p u_g = u_{p0}^2$
t = L/u_g (s) travel time through guide of length L
Higher modes have lower u_g ⇒ arrive later.

BOUNDARY

η_i, θ_t — imp. (Ω)
Γ, τ, R, T — coeffs (unitless)
τ = 1 + Γ; R = |Γ|²
k_i = ω√ε_{r,i}/c (rad/m)
α₂ (Np/m), β₂ (rad/m): lossy
z = 0 — boundary plane (m)
η₀ = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ_i, θ_t — angles from normal
n_i = √ε_{r,i} — index of refr.
t_i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k+normal
⊥: **E** ⊥ this plane
||: **E** || this plane

WAVEGUIDE

a, b — wide, narrow dim. (m);
a > b
m, n — mode indices (int.)
f_{mn} — cutoff (Hz)
β — prop. const (rad/m)
Z_{TE}/TM — wave imp. (Ω)
E_x — E-field comp. (V/m)
H_y — H-field comp. (A/m)
η = η₀/√ε_r

VELOCITIES

u_{p0} = c/√ε_r — bulk phase vel.
u_p = u_{p0}/√[1 - (f_c/f)²]
u_g = u_{p0}√[1 - (f_c/f)²]
u_pu_g = u_{p0}²
t = L/u_g — travel time
c = 3 × 10⁸ m/s

POWER & UNITS

%P_r, %P_t — pwr refl./trans. (unitless)
%P_r = 100|Γ|²
%P_t = 100|τ|² η₁/η₂
%P_r + %P_t = 100
σ (S/m), ε (F/m), μ (H/m)
ε₀ = 8.85 × 10⁻¹² F/m
μ₀ = 4π × 10⁻⁷ H/m

DIPOLE TYPE BY LENGTH
l = antenna rod length (m)

 λ (wavelength) = $\frac{c}{f} = \frac{3 \times 10^8}{f}$ (m)		
<i>l</i> /λ	Type	Use formula
< 1/50	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D=1.5$
= 1/4	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
= 1/2	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73\,\Omega$
= 1	Full-wave	ARB, $D=2.41$, $R_{\text{rad}}=199\,\Omega$
other	Arbitrary	ARB formula

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors <i>small l</i> ≪ λ $\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} \frac{e^{-jkR}}{R} \sin\theta$, $\vec{H}_\phi = \vec{E}_\theta/\eta_0$ Power density formulas <i>l</i> /λ < 1/50; <i>far field</i>	
Power density	$S(R, \theta) = S_{\text{max}} \sin^2\theta$ (W/m ²) $S_{\text{max}} = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}$ (W/m ²)
Max pwr density	
<i>l</i> : rod length (m)	<i>I</i> ₀ : peak current (A)
<i>R</i> : obs. distance (m)	<i>θ</i> : angle from dipole axis (rad)
<i>k</i> =2π/λ (rad/m)	<i>η</i> ₀ =120π≈377 Ω
Total radiated power	$P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2$ (W)

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at <i>R</i> any <i>l</i>
$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$
<i>At θ = 0, π: bracket is 0/0. L'Hôpital ⇒ S = 0 (no axial radiation). TI-36X Pro alt: evaluate bracket at θ = 0.001 rad → tiny → 0.</i>
Normalized pattern dimensionless $F(\theta)=S(\theta)/S_{\text{max}}$

Antenna formulas usually take *θ* in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

<i>N</i> identical elements along axis, spacing <i>d</i> , equal phase (broadside, <i>θ</i> _{max} = π/2). Total pattern <i>S</i> = <i>S_e</i> · <i>F_a</i> (<i>S_e</i> = single elem).
Array factor (uniform, broadside) $k = 2\pi/\lambda$, $\gamma = kd \cos\theta$ $F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$; $F_a^{\text{max}} = N^2$ at <i>θ</i> = π/2
Normalize $F_a^{\text{norm}} = F_a/N^2$
HPBW (broadside, uniform) use <i>Nd</i>, not (<i>N</i>−1)<i>d</i> $\beta \approx 0.88 \frac{\lambda}{Nd}$ (rad) = 50.8° · $\frac{\lambda}{Nd}$ <i>Grating lobes appear if d > λ. Avoid.</i>

BEAM PARAMETERS — β, Ω_p, *D*

Normalize always start here $F(\theta) = S(\theta)/S_{\text{max}}$ (unitless) Beamwidth β at threshold <i>X</i> Set $F(\theta_X) = X$, solve for <i>θ_X</i> ; β = 2 <i>θ_X</i> (symmetric)		
Threshold	<i>X</i> = 10 ^{dB/10}	<i>θ_X</i> (Gauss. short-cut)
HPBW (−3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
−6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
−10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$	any <i>X</i>	$F(\theta_X) = X$
Pattern solid angle Ω _p recognize pattern, look up $\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin\theta \, d\theta \, d\phi$ (sr) <i>No φ dep.:</i> Ω _p = 2π∫ ₀ ^π <i>F</i> (<i>θ</i>) sin <i>θ</i> <i>dθ</i> . <i>θ is the integration variable — DON'T plug in a number for θ (e.g. θ_{1/2}). Integrate over θ ∈ [0, π].</i>		
Pattern <i>F</i> (<i>θ</i>)	Ω _p (sr)	
Isotropic (<i>F</i> = 1)	4π ≈ 12.57	
Hertzian sin ² <i>θ</i>	8π/3 ≈ 8.38	
Half-wave dipole	≈ 7.66	
Narrow Gauss <i>e</i> ^{−<i>aθ</i>²}	π/ <i>a</i> = π <i>θ</i> _{1/2} ² /ln 2	
2-axis HPBW (aperture)	β _{<i>xz</i>} · β _{<i>yz</i>}	numerical
Other (use integral)	numerical	

TI-36X Pro: [2nd] [d/dx] for $\int_{\square}^{\square}$, RAD mode, multiply by 2π.

Directivity <i>D</i> always <i>D</i> = 4π/Ω_p; common shortcuts below $D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$
WORD TRAP: “direction” = <i>θ</i> _{max} (rad); “directivity” = <i>D</i> (unitless). Dipoles → COMMON DIPOLE PARAMS table. Aperture: <i>D</i> ≈ 4π <i>A_p</i> /λ ² . 2-axis HPBW: <i>D</i> ≈ 4π/(β _{<i>xz</i>} β _{<i>yz</i>}). Circular: <i>D</i> ≈ 4π/β ² . Gain (if asked) $\xi = \text{rad. eff.}$, 0 < ξ ≤ 1 $G = \xi D$ $G_{\text{dB}} = 10 \log_{10} G$ Lossless / ideal antenna ⇒ <i>G</i> = <i>D</i> .

COMMON DIPOLE PARAMETERS

Lossless ($\xi = 1$): $G = D$.				
Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
Monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199
Half-wave: $P_{\text{rad}} = 36.6 I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but P_{rad} , R_{rad} halved.				
ANTENNA AREAS (A_e , A_p)				
Effective area antenna's RX cross section; D from BEAM PARAMS or table above				
$A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)				
Physical cross section wire dipole, diameter d ; l see table above				
$A_p = l d$ (m ² , any dipole)				
Inverse: D from A_e				
$D = \frac{4\pi A_e}{\lambda^2}$ (unitless)				
Thin wire: $A_e \gg A_p$ (e.g., Hertzian $A_e = 3\lambda^2/8\pi \approx 0.119\lambda^2$).				

FRIIS TRANSMISSION FORMULA

For each antenna's G : if given by NAME ("half-wave dipole" etc.) → COMMON DIPOLE PARAMS ($G = D$, lossless). Else (dB/linear value given) → use as given (convert dB → linear).

Pwr density at RX, then RX pwr G_t, G_r linear gains;
 $\lambda = c/f$

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{); } P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t \text{ (W)}$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) when patterns NOT peak-aligned

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert any dB gain to linear: $G = 10^{G_{\text{dB}}/10}$.

3. Get G_t, G_r by antenna type:

- named dipole → COMMON DIPOLE PARAMS ($G = D$ lossless)
- aperture / dish → APERTURE ANTENNAS ($G = D \approx 4\pi A_p / \lambda^2$)
- arbitrary pattern $F(\theta) \rightarrow$ BEAM PARAMS ($G = \xi \cdot 4\pi / \Omega_p$)
- given in dB → dB ↔ LINEAR table

4. Plug into Friis.

Use linear G , not dB. Convert FIRST.

DIPOLE

l — antenna length (m)
 $\lambda = c/f$ (m); $c = 3 \times 10^8$ m/s
 I_0 — peak current (A)
 $k = 2\pi/\lambda$ (rad/m)
 R — obs. dist. (m); θ from axis
 $\eta_0 = 120\pi \approx 377 \Omega$
 $S(R, \theta)$ — pwr density (W/m²)

BEAM

$F(\theta) = S/S_{\max}$ (unitless)
 a — coef. on θ^2 in Gaussian
 β — HPBW (rad or °)
 Ω_p — pattern solid angle (sr)
 $D = 4\pi/\Omega_p$ — directivity
 ξ rad. eff.; $G = \xi D$
 $D_{\text{dB}} = 10 \log_{10} D$; $D = 10^{D_{\text{dB}}/10}$

FRIIS / dB

P_t — power transmitted (W)
 P_r — power received (W)
 G_t — transmit gain (linear)
 G_r — receive gain (linear)
 $S_r = G_t P_t / (4\pi R^2)$ (W/m²)
 $P_r = G_t G_r (\lambda/4\pi R)^2 P_t$
 $G = 10^{G_{\text{dB}}/10}$
 $3 \text{ dB} = \times 2, 10 \text{ dB} = \times 10$
 $P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m²)
 $A_e = \lambda^2 D / (4\pi)$ (m²)
 $A_e \approx A_p$ for aperture
 $D \approx 4\pi A_p / \lambda^2$ (unitless)
 $\beta \approx 0.88\lambda/\ell$ (rect/sq.)
 $\beta \approx 1.02\lambda/d$ (circular)
 $2A_p \Rightarrow 2D, \beta/\sqrt{2}$

ARRAY

N — number of elements
 d — spacing (m)
 $\gamma = kd \cos \theta$
 $F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$
 $F_a^{\max} = N^2$ at $\theta = \pi/2$
 $F_a^{\text{norm}} = F_a / N^2$
 $\beta \approx 0.88\lambda/(Nd)$ (broadside)